

ECO 146 Math Toolkit

Formula Sheet for Applied Microeconomic Analysis
Summer 2 2026

1. Algebra, Lines, and Graphs

Linear equations

$$Y = a + bX$$

$$\text{slope} = b = \frac{\Delta Y}{\Delta X}$$

$$\text{vertical intercept} = a \quad \text{when } X = 0$$

Demand and supply forms

$$Q_D = a - bP \quad (b > 0)$$

$$Q_S = c + dP \quad (d > 0)$$

$$P = \frac{a}{b} - \frac{1}{b}Q_D \quad (\text{inverse demand})$$

$$P = -\frac{c}{d} + \frac{1}{d}Q_S \quad (\text{inverse supply})$$

Equilibrium

$$Q_D = Q_S$$

$$P^* = \text{equilibrium price}, \quad Q^* = \text{equilibrium quantity}$$

Solving two linear equations

$$\begin{cases} ax + by = A \\ cx + dy = B \end{cases}$$

substitution or elimination

2. Revenue, Cost, Profit, and Averages

Revenue

$$TR = P \times Q$$

$$TR(P) = P \cdot Q(P)$$

$$TR(Q) = P(Q) \cdot Q$$

$$MR(Q) = \frac{dTR(Q)}{dQ}$$

$$AVC(Q) = \frac{VC(Q)}{Q}$$

$$ATC(Q) = \frac{TC(Q)}{Q} = AFC(Q) + AVC(Q)$$

$$MC(Q) = \frac{dTC(Q)}{dQ}$$

Cost

$$TC(Q) = FC + VC(Q)$$

$$AFC(Q) = \frac{FC}{Q}$$

Profit

$$\pi(Q) = TR(Q) - TC(Q)$$

$$\frac{d\pi}{dQ} = MR(Q) - MC(Q)$$

profit maximization: $MR = MC$

3. Derivatives and Marginal Analysis

Marginal interpretation

Marginal = additional change from one more unit

$$MB(Q) = \frac{dB(Q)}{dQ}, \quad MC(Q) = \frac{dC(Q)}{dQ}$$

optimal choice: $MB = MC$

Basic derivative rules

$$\frac{d}{dx}(c) = 0$$

$$\frac{d}{dx}(ax) = a$$

$$\begin{aligned}\frac{d}{dx}(x^n) &= nx^{n-1} \\ \frac{d}{dx}(ax^n) &= anx^{n-1} \\ \frac{d}{dx}[f(x) + g(x)] &= f'(x) + g'(x) \\ \frac{d}{dx}[f(x) - g(x)] &= f'(x) - g'(x)\end{aligned}$$

Useful optional rules

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$

4. Optimization

Unconstrained optimization

$$\max_x f(x)$$

$$\text{FOC: } f'(x) = 0$$

$$\text{maximum check: } f''(x) < 0$$

$$\text{minimum check: } f''(x) > 0$$

Common economic forms

$$\max_Q \pi(Q) = TR(Q) - TC(Q)$$

$$\frac{d\pi}{dQ} = 0 \iff MR = MC$$

$$\max_Q [B(Q) - C(Q)]$$

$$\frac{dB}{dQ} = \frac{dC}{dQ} \iff MB = MC$$

5. Partial Derivatives and Multivariable Functions

Partial derivatives

$$f(x, y)$$

$\frac{\partial f}{\partial x}$: derivative with respect to x holding y constant

$\frac{\partial f}{\partial y}$: derivative with respect to y holding x constant

Power-product form

$$f(x, y) = Ax^a y^b$$

$$\frac{\partial f}{\partial x} = Aax^{a-1}y^b$$

$$\frac{\partial f}{\partial y} = Abx^a y^{b-1}$$

Consumer theory

$$U(x, y) = \text{utility}$$

$$MU_x = \frac{\partial U}{\partial x}, \quad MU_y = \frac{\partial U}{\partial y}$$

$$\text{slope of indifference curve} = -\frac{MU_x}{MU_y}$$

$$MRS_{xy} = \frac{MU_x}{MU_y} \quad (\text{magnitude})$$

$$\text{budget constraint: } P_x x + P_y y = I$$

$$\text{interior optimum: } \frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

$$\text{equivalently: } MRS_{xy} = \frac{P_x}{P_y}$$

Production theory

$$Q = F(L, K)$$

$$MP_L = \frac{\partial Q}{\partial L}, \quad MP_K = \frac{\partial Q}{\partial K}$$

$$\text{slope of isoquant} = -\frac{MP_L}{MP_K}$$

$$MRTS_{LK} = \frac{MP_L}{MP_K} \quad (\text{magnitude})$$

6. Elasticities

General elasticity

$$E_{Y,X} = \frac{\% \Delta Y}{\% \Delta X}$$

$$E_{Y,X} = \frac{dY}{dX} \cdot \frac{X}{Y}$$

Price elasticity of demand

$$E_D = \frac{\% \Delta Q_D}{\% \Delta P}$$

$$E_D = \frac{dQ_D}{dP} \cdot \frac{P}{Q_D}$$

$$|E_D| > 1 : \text{elastic}$$

$$|E_D| < 1 : \text{inelastic}$$

$$|E_D| = 1 : \text{unit elastic}$$

Other elasticities

$$E_S = \frac{dQ_S}{dP} \cdot \frac{P}{Q_S}$$

$$E_I = \frac{dQ}{dI} \cdot \frac{I}{Q}$$

$$E_{xy} = \frac{dQ_x}{dP_y} \cdot \frac{P_y}{Q_x}$$

Arc elasticity

$$E_D^{arc} = \frac{\Delta Q}{\Delta P} \cdot \frac{\bar{P}}{\bar{Q}}$$

$$\bar{P} = \frac{P_1 + P_2}{2}, \quad \bar{Q} = \frac{Q_1 + Q_2}{2}$$

$$E_D^{arc} = \frac{Q_2 - Q_1}{P_2 - P_1} \cdot \frac{P_1 + P_2}{Q_1 + Q_2}$$

7. Welfare and Policy Geometry

Areas

rectangle area = base $\times$ height

triangle area = $\frac{1}{2} \times$ base $\times$ height

Consumer and producer surplus

CS = area below demand and above price

PS = area above supply and below price

$$TS = CS + PS$$

Price controls

shortage = $Q_D - Q_S$ if $P_{max} < P^*$

surplus = $Q_S - Q_D$ if $P_{min} > P^*$

Per-unit tax

$$P_B - P_S = t$$

tax revenue = $t \times Q_{tax}$

$$DWL = \frac{1}{2}t(Q^* - Q_{tax})$$

8. Common Decision Rules

$MB > MC \Rightarrow$ do more

$MB < MC \Rightarrow$ do less

$MB = MC \Rightarrow$ optimal interior choice

$MR > MC \Rightarrow$ increase output

$MR < MC \Rightarrow$ decrease output

$MR = MC \Rightarrow$ profit-maximizing output

$Q_D > Q_S \Rightarrow$ shortage

$Q_S > Q_D \Rightarrow$ surplus